UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH
Escola Superior d'Enginyeries Industrial,
Aeroespacial i Audiovisual de Terrassa

Assignment 2 - Solving Kepler equation

Leoni Fritsch, Martí Güell, Gerard Martínez

Advanced Astrodynamics

ABSTRACT

This report analyzes the convergence behavior of three numerical methods for solving Kepler's equation, namely the Newton–Raphson, Fixed-Point Iteration, and Halley's methods. Each method is tested using three different initial guesses, and their convergence properties are assessed over a wide range of eccentricities and mean anomalies.

1 INTRODUCTION

Kepler's equation is a fundamental relation in celestial mechanics, connecting the mean anomaly M, the eccentric anomaly E, and the eccentricity e through:

$$M = E - e\sin E. \tag{1}$$

Since this equation cannot be solved analytically for E, iterative numerical methods are required.

2 NUMERICAL METHODS

Three iterative methods were implemented and analyzed: Newton–Raphson, Fixed-Point, and Halley's method. Each is briefly summarized below.

2.1 Fixed-Point Iteration

A simpler approach consists of rearranging Kepler's equation as:

$$E = M + e\sin E, \tag{2}$$

and using the iteration:

$$E_{n+1} = M + e\sin E_n. \tag{3}$$

Although easy to implement, convergence is guaranteed only when $e < 0.66$, and the rate is linear.

2.2 Newton–Raphson Method

The Newton–Raphson (NR) method is a second-order iterative technique that updates the solution as:

$$E_{n+1} = E_n - \frac{E_n - e\sin E_n - M}{1 - e\cos E_n}. \tag{4}$$

The method generally converges quadratically near the solution if the initial guess is close enough.

2.3 Halley's Method

Halley's method extends Newton–Raphson to third-order convergence. The iteration formula is:

$$E_{n+1} = E_n - \frac{f(E_n)}{f'(E_n) - \frac{1}{2} f(E_n) f''(E_n)/f'(E_n)}, \tag{5}$$

where

$$f(E) = E - e\sin E - M, \quad f'(E) = 1 - e\cos E, \quad f''(E) = e\sin E.$$

This method typically converges faster than NR, particularly for high eccentricities.

3 METHODOLOGY

The analysis was performed using Python, testing each solver over a grid of mean anomalies $M \in [0, 2\pi]$ and eccentricities $e \in [0, 1)$. The number of iterations required to achieve a tolerance of 10^{-12} was recorded.

Three different initial guesses were evaluated:

1. $E_0 = M$
2. $E_0 = \pi$
3. $E_0 = M + e\cos M$

4 RESULTS AND DISCUSSION

As can be seen in Figure 1, a comparison between methods has been done, using e = 0.95 and M = 350º. Newton and Halley's methods are the fastest ones, converging in less than 10 iterations. Meanwhile, the fixed point method goes up to almost 50 iterations.

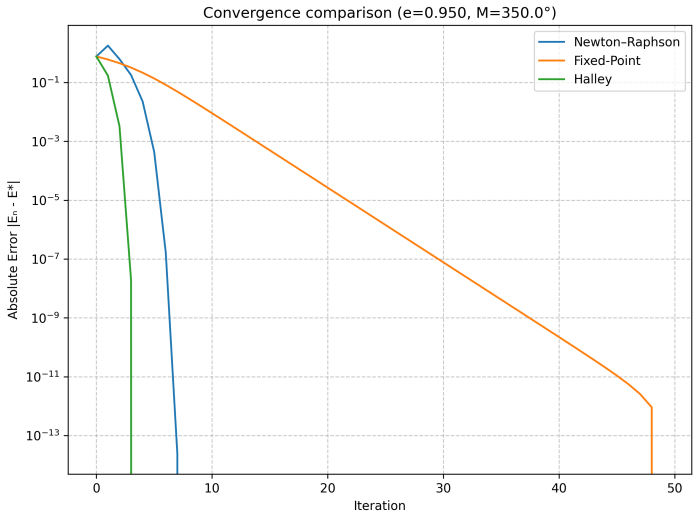
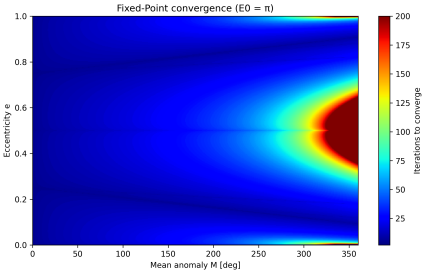


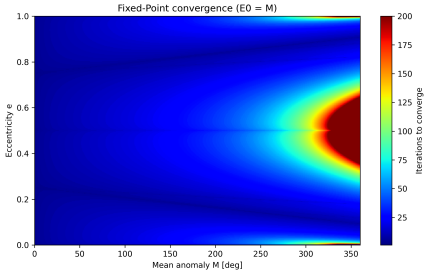
Figure 1. Comparison of convergence rates among the three methods.

Figures 2a–4c show the convergence maps and iteration histories for all methods. The Newton–Raphson and Halley's methods converge rapidly for all eccentricities except near $e \approx 1$ and $M \approx 0$. The fixed-point iteration diverges for medium excentricities and near $e \approx 0, 1$, confirming its limited applicability.

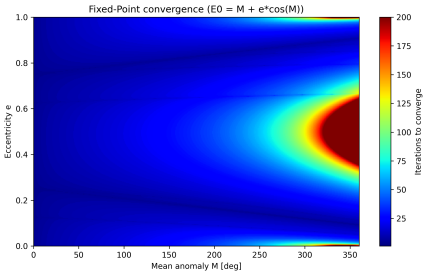
Overall, the method that seems to have presented the best results is, as expected, the Halley's method, for being a third order convergence approximation.



(a) Convergence map for fixed-point method using $E_0 = \pi$.

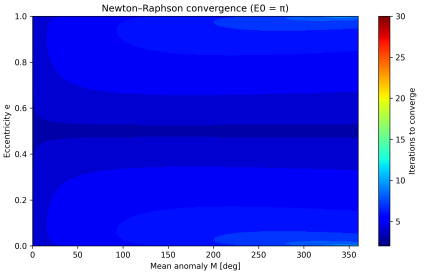


(b) Convergence map for fixed-point method using $E_0 = M$.

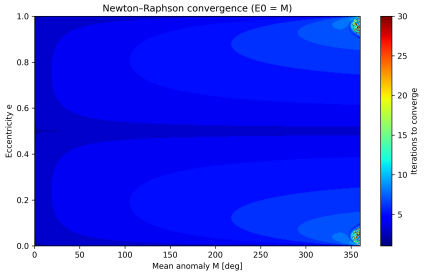


(c) Convergence map for fixed-point method using $E_0 = M + e\cos(M)$.

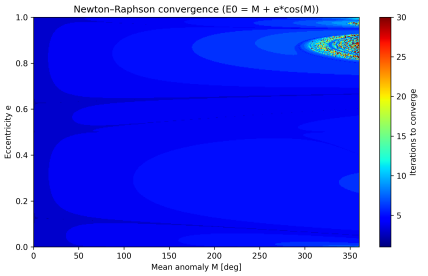
Figure 2. Convergence maps for the fixed-point method using different initial guesses E_0.



(a) Convergence map for Newton–Raphson method using $E_0 = \pi$.

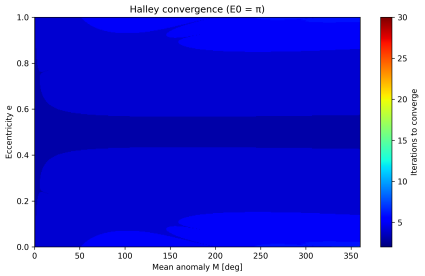


(b) Convergence map for Newton–Raphson method using $E_0 = M$.

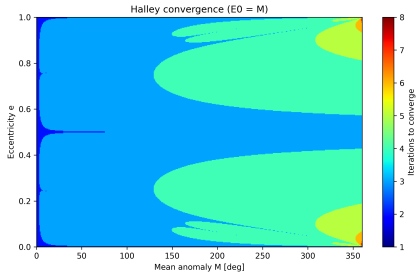


(c) Convergence map for Newton–Raphson method using $E_0 = M + e\cos(M)$.

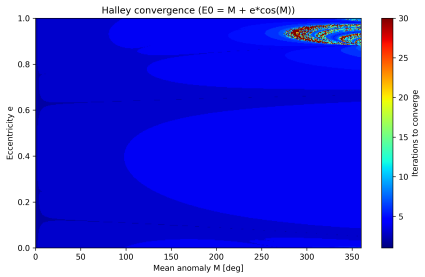
Figure 3. Convergence maps for the Newton–Raphson method using different initial guesses E_0.



(a) Convergence map for Halley method using $E_0 = \pi$.



(b) Convergence map for Halley method using $E_0 = M$.



(c) Convergence map for Halley method using $E_0 = M + e\cos(M)$.

Figure 4. Convergence maps for the Halley method using different initial guesses E_0.

5 CONCLUSION

This study compared the convergence performance of three iterative methods—Fixed-Point Iteration, Newton–Raphson, and Halley's method—for solving Kepler's equation across a wide range of eccentricities and mean anomalies.

The results clearly show that the Fixed-Point Iteration method is simple but limited, and exhibits slow, linear convergence. The Newton–Raphson method, on the other hand, provides a robust and efficient solution for most cases, achieving quadratic convergence with minimal computational effort. It consistently performs well across the full range of tested parameters when a reasonable initial guess is chosen.

The Halley's method demonstrates the fastest convergence, as expected from its third-order nature. While it incurs slightly higher computational cost per iteration due to second-derivative evaluations, its superior convergence rate makes it particularly advantageous for high-eccentricity or challenging initial conditions.

Overall, the Newton–Raphson method offers the best compromise between simplicity, efficiency, and robustness, making it the preferred choice for general-purpose applications. Halley's method is recommended when maximum precision or rapid convergence is required, while the Fixed-Point approach remains suitable only for educational purposes.